2.4.5 "Processed" Methods and Conjugacy

One of the intriguing ideas that can be exploited to improve numerical integrators is that of *conjugacy*. In general, we say that two maps $\boldsymbol{A}$ and $\boldsymbol{B}$ are conjugate if there is a homeomorphism[7] $\boldsymbol{\chi}$ such that

$$\boldsymbol{A} = \boldsymbol{\chi}^{-1} \circ \boldsymbol{B} \circ \boldsymbol{\chi}.$$

That is, we may evaluate one map by first transforming the input, then applying the other map, then transforming the output by the inverse of the first transformation. Conjugate maps have the property that their iterates are also conjugate, since

$$\begin{aligned} \boldsymbol{A}^n &= (\boldsymbol{\chi}^{-1} \circ \boldsymbol{B} \circ \boldsymbol{\chi})^n \\ &= (\boldsymbol{\chi}^{-1} \circ \boldsymbol{B} \circ \boldsymbol{\chi}) \circ (\boldsymbol{\chi}^{-1} \circ \boldsymbol{B} \circ \boldsymbol{\chi}) \cdots (\boldsymbol{\chi}^{-1} \circ \boldsymbol{B} \circ \boldsymbol{\chi}) \\ &= \boldsymbol{\chi}^{-1} \circ \boldsymbol{B}^n \circ \boldsymbol{\chi}. \end{aligned}$$

If $\boldsymbol{A}$ and $\boldsymbol{B}$ are maps of phase space, then the conjugacy implies that they have equivalent stability properties under iteration, since if $\boldsymbol{B}^n(z_0) \to z^*$, as $n \to \infty$, for all initial points z_0, then also $\boldsymbol{A}^n(z_0) \to \boldsymbol{\chi}^{-1}(z^*)$.

A common use made of conjugacy in the setting of numerical integration is to increase the order of accuracy. Numerical integrators are dependent on the stepsize h. The idea is to introduce a conjugacy via a map χ_h which also depends on the stepsize and by judicious design of the map, to eliminate leading terms in the expansion of the local error of the method. As an illustration, the Symplectic Euler method turns out to be conjugate to the Verlet method (see Exercise 12). One sometimes refers to the "effective order" of a numerical method as the order attainable via processing, thus the effective order of the Symplectic Euler method would be two.

Let us suppose that we have such a conjugacy between two numerical methods $\mathscr{G}_h$ and $\widetilde{\mathscr{G}}_h$, that is

$$\mathscr{G}_h = \boldsymbol{\chi}_h^{-1} \circ \widetilde{\mathscr{G}}_h \circ \boldsymbol{\chi}_h,$$

defined in such a way that $\mathscr{G}_h$ has order r and $\widetilde{\mathscr{G}}_h$ has order s where $s < r$. Then, given an initial condition z_0, we first modify ("pre-process") this to $\tilde{z}_0 = \boldsymbol{\chi}_h(z_0)$, then take multiple steps with the method $\widetilde{\mathscr{G}}_h$, and finally transform ("post-process") the end result by $\boldsymbol{\chi}_h^{-1}$. The resulting approximation will be of order r even though all timestepping was performed using a lower order method. The situation is diagrammed in Fig. 2.7.

[7] A homeomorphism is a continuous bijection which has a continuous inverse.